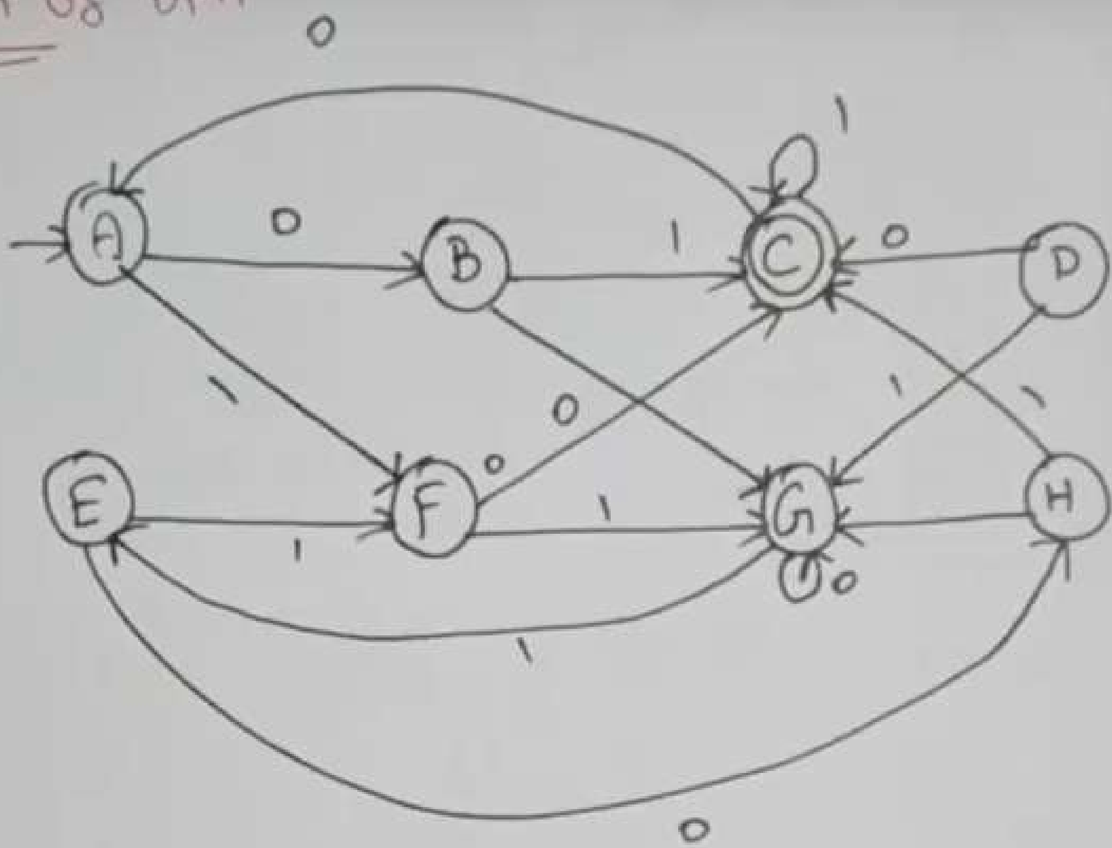
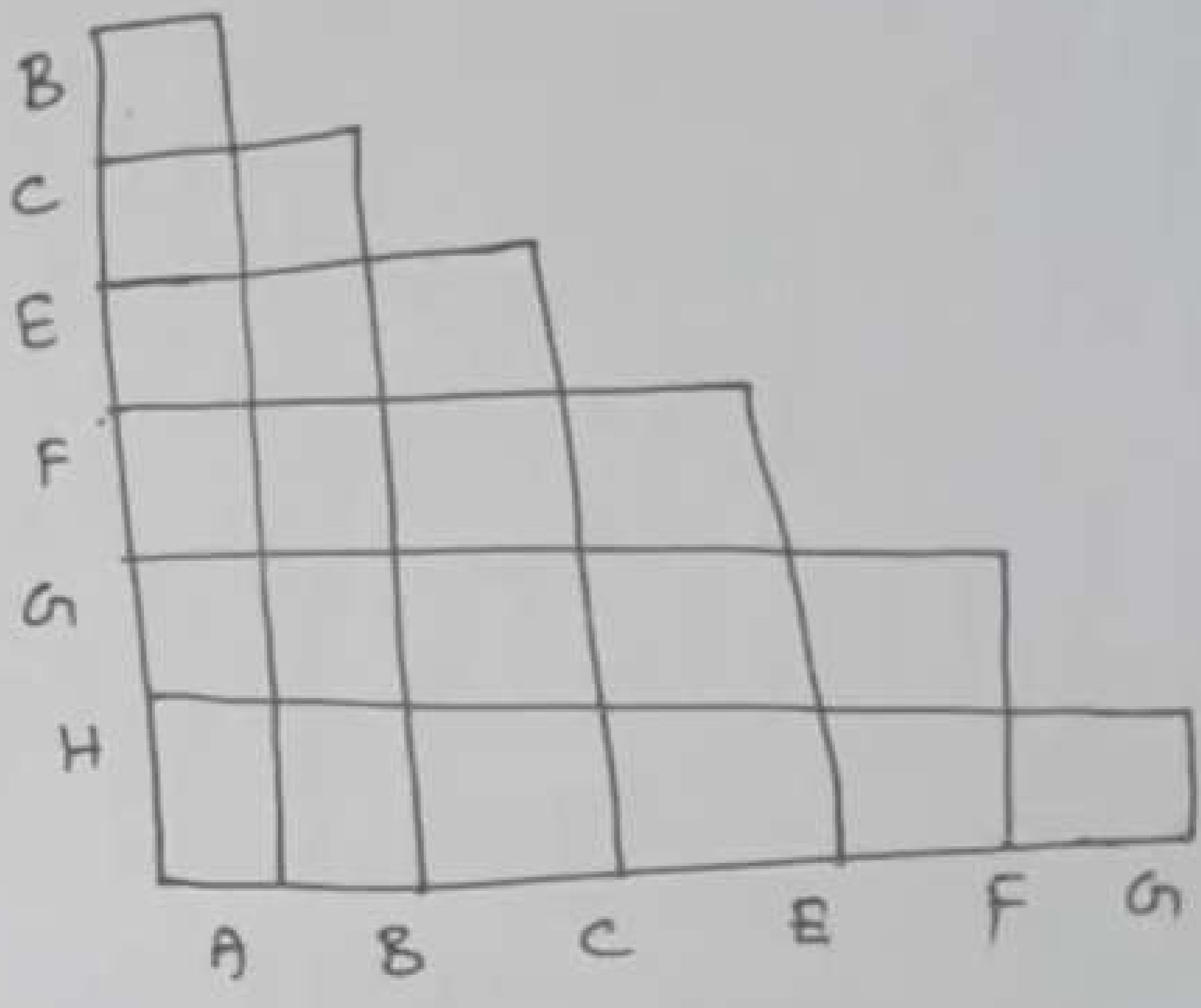
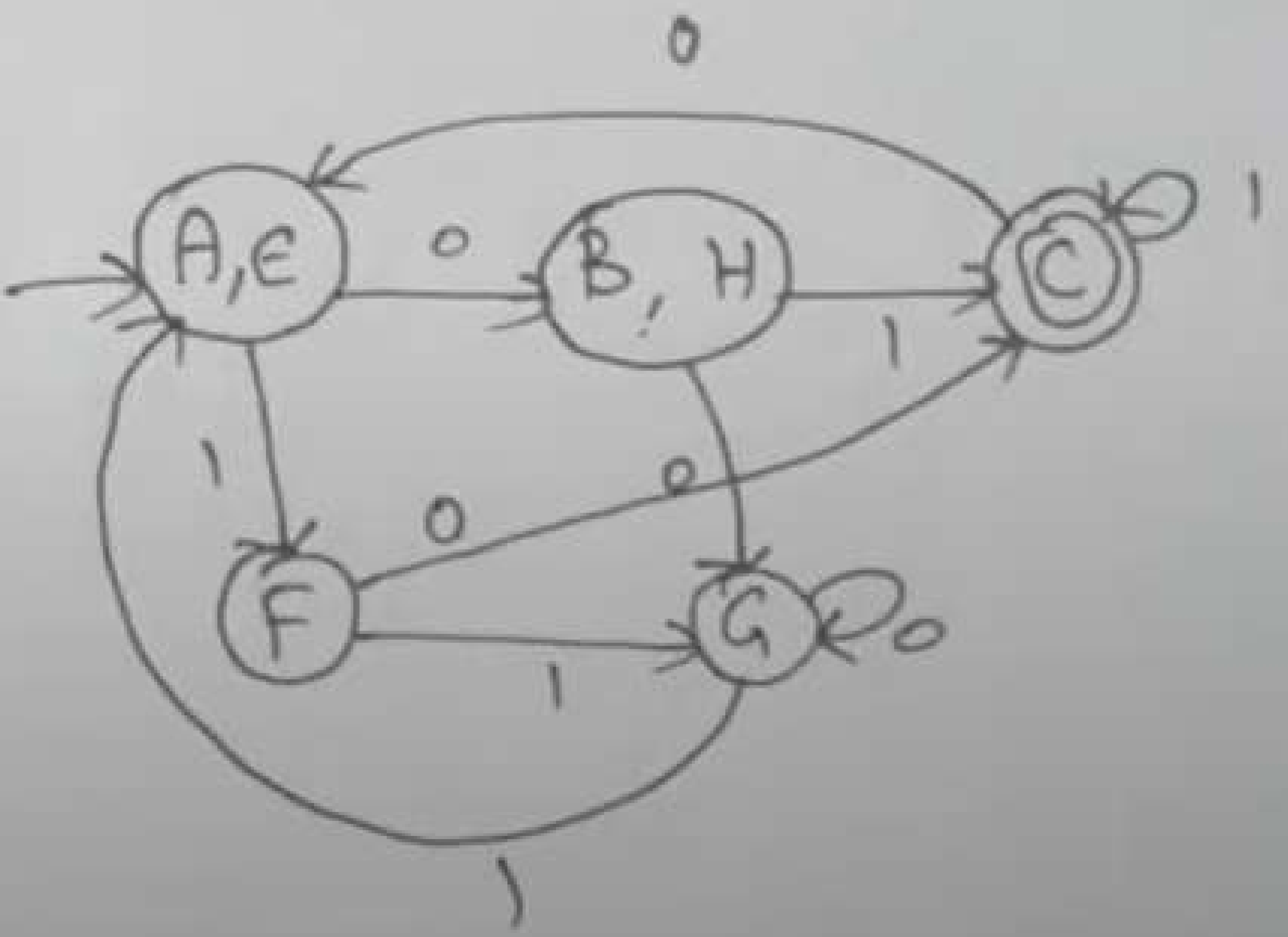
Minimization of DFA




B
C
E
F
G
H
A
B
C
E
F
G

| | 0 | 1 |
|---|---|---|
| A | B | F |
| B | G | C |
| C | A | C |
| E | H | F |
| F | C | G |
| G | G | E |
| H | G | C |

| | A | B | C | E | F | G |
|---|---|---|---|---|---|---|
| B | X | | | | | |
| C | X | X | | | | |
| E | | X | X | | | |
| F | X | X | X | X | | |
| G | | X | X | | X | |
| H | X | | X | X | X | X |



0
A,E
0
B,H
C
1
1
1
0
0
F
G
0
1
1

# Minimization of DFA - Table Filling Method

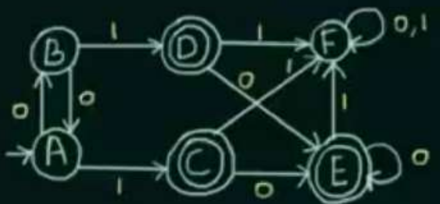
(Myhill-Nerode Theorem)


Steps:

1) Draw a table for all pairs of states (P,Q)
2) Mark all pairs where $P \in F$ and $Q \notin F$
3) If there are any Unmarked pairs (P,Q) such that $[\delta(P,x), \delta(Q,x)]$ is marked, then mark [P,Q] where 'x' is an input symbol
   REPEAT THIS UNTIL NO MORE MARKINGS CAN BE MADE
4) Combine all the Unmarked Pairs and make them a single state in the minimized DFA

# Minimization of DFA - Table Filling Method

(Myhill-Nerode Theorem)




| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| A | | | | | | |
| B | | | | | | |
| C | ✓ | ✓ | | | | |
| D | ✓ | ✓ | | | | |
| E | ✓ | ✓ | | | | |
| F | | | ✓ | ✓ | ✓ | |

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(P,Q)

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| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| A | | | | | | |
| B | | | | | | |
| C | ✓ | ✓ | | | | |
| D | ✓ | ✓ | | | | |
| E | ✓ | ✓ | | | | |
| F | ✓ | ✓ | ✓ | ✓ | ✓ | |

$$(D,C) - \left.\begin{array}{l}\delta(D,0) = E\\ \delta(C,0) = E\end{array}\right\} \left.\begin{array}{l}\delta(D,1) = F\\ \delta(C,1) = F\end{array}\right\}$$

$$(E,C) - \left.\begin{array}{l}\delta(E,0) = E\\ \delta(C,0) = E\end{array}\right\} \left.\begin{array}{l}\delta(E,1) = F\\ \delta(C,1) = F\end{array}\right\}$$

$$(E,D) - \left.\begin{array}{l}\delta(E,0) = E\\ \delta(D,0) = E\end{array}\right\} \left.\begin{array}{l}\delta(E,1) = F\\ \delta(D,1) = F\end{array}\right\}$$

$$\underline{(F,A)} - \left.\begin{array}{l}\delta(F,0) = F\\ \delta(A,0) = B\end{array}\right\} \left.\begin{array}{l}\delta(F,1) = F\\ \delta(A,1) = C\end{array}\right\}$$

$$\underline{(F,B)} - \left.\begin{array}{l}\delta(F,0) = F\\ \delta(B,0) = A\end{array}\right\}$$

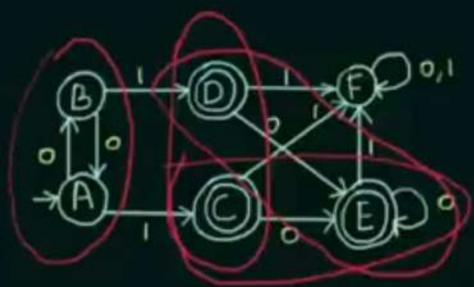
$$(B,A) - \left.\begin{array}{l}\delta(B,0) = A\\ \delta(A,0) = B\end{array}\right\} \left.\begin{array}{l}\delta(B,1) = D\\ \delta(A,1) = C\end{array}\right\}$$

$\delta(D,0) = E \}$ $\delta(D,1) = F \}$

$\delta(B,0) = A \}$

$(B,A) - \left.\begin{matrix}\delta(B,0) = A \\ \delta(A,0) = B\end{matrix}\right\} \left.\begin{matrix}\delta(B,1) = D \\ \delta(A,1) = C\end{matrix}\right\}$

$(A,B)$ $(D,C)$ $(E,C)$ $(E,D)$



$\delta(D,0) = E \}$ $\delta(D,1) = F \}$ $\delta(B,0) = A \}$

$(B,A)$ - $\left.\begin{array}{l}\delta(B,0) = A \\ \delta(A,0) = B\end{array}\right\}$ $\left.\begin{array}{l}\delta(B,1) = D \\ \delta(A,1) = C\end{array}\right\}$

$(A,B)$ $(D,C)$ $(E,C)$ $(E,D)$